HackStellar · PS-19

Complaint Compass

Every complaint, from any channel, filed under the right problem.

Kartik · Jnanottam



The problem

Complaints arrive everywhere, worded every way.

Email
Money debited, no order

Chat
paise kat gaye lekin order nahi bana

Call
Paid by UPI, the app shows nothing

Social
@basketnow where is my ₹415??

Chat
Same customer, third message today

4
channels, each its own format

100s
of wordings for one problem

1
customer, not three, across channels

Every complaint, the same few questions

Four choices file a complaint.

1

Who sent it?

→ One person on three channels counts once

2

A complaint?

→ Praise, spam, questions: set aside

3

About what?

→ Which of 7 areas

→ How serious

4

Known problem?

→ Yes: filed

→ Unsure: checked closer

→ New: waits for 3 customers

→ Money or safety: urgent now

Every step is a choice, not writing.

Live demo

Type a complaint. Watch where it stops.

“paise kat gaye lekin order nahi bana, please check”

→ Money debited but order not placed · Hinglish

“I was charged twice for one order since I updated the app”

→ Duplicate payment debits after app update

“Love the 10 minute delivery, great job team!”

→ Set aside: not a complaint



Be the customer

hackstellar-three.vercel.app/submit

The usual way

An LLM reads every complaint.



Slow

Every complaint waits for an answer to be written.

Expensive

It writes for every complaint. Output tokens cost the most.

Same job

Against an LLM reading every complaint.

65–85%
cheaper

96%
fewer output tokens

2.3×
faster

Our side measured with the real models on our test run. The LLM-everywhere side is projected from calibrated token counts, not run.

Our idea

Choose, don’t write.

LLM a writer

“This complaint appears to concern a failed payment where the user…”

Writes for every complaint **slow · costly**

Jev a decision model

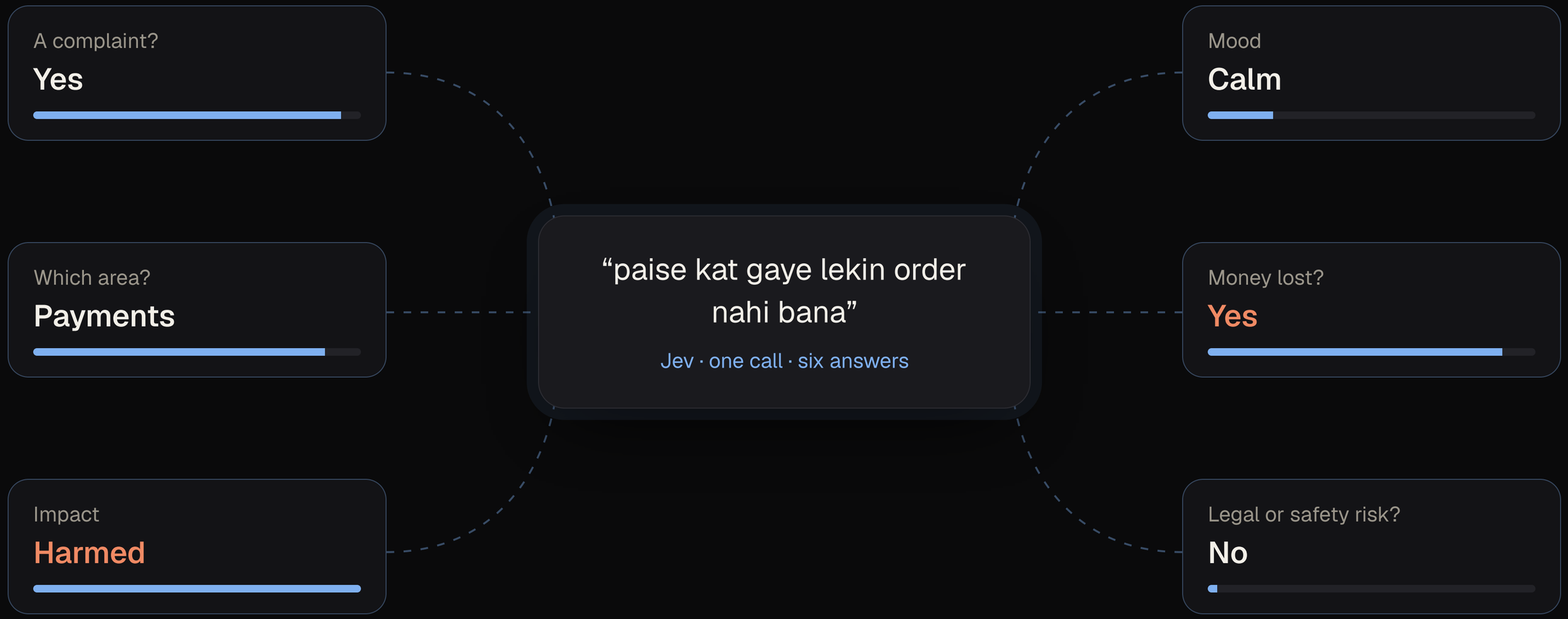


Picks from options, writes nothing **fast · cheap**

The LLM writes only when something new needs a name.

The gate

One call. Six answers.



Who does what

Three workers, priced by what they do.

Jev

decision model · picks, never writes

- Gate: 6 questions
- Verify: same problem?
- Group check: known in new words?

Free

runs on our machine

- Read 4 channels
- Find the customer
- Match by meaning
- Waiting list
- Ranking

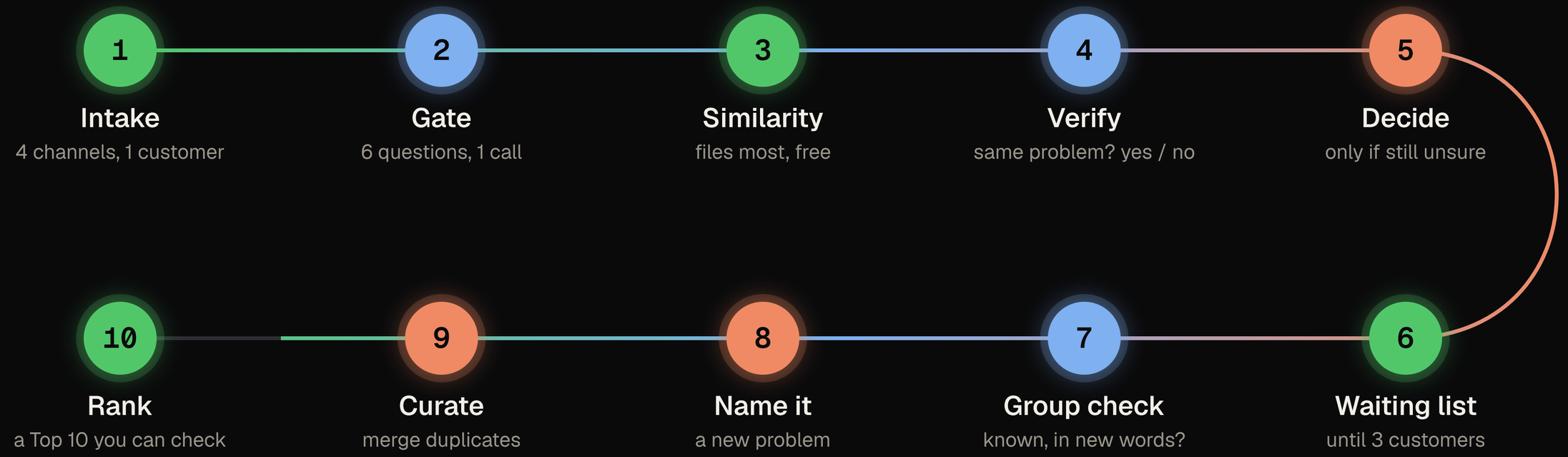
LLM

Claude Haiku · only where words are needed

- The last unsure few
- Name new problems
- Merge duplicates

The whole flow

Ten steps, cheapest first.



Tested live · real models · an answer key

300 new complaints. One mistake.

Batch 1 · 50 complaints

49/50

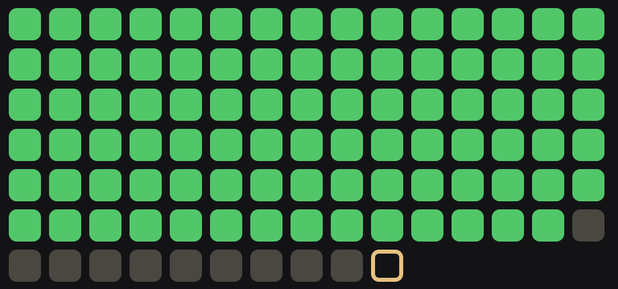
right · 1 missed



Batch 2 · 100 complaints

99/100

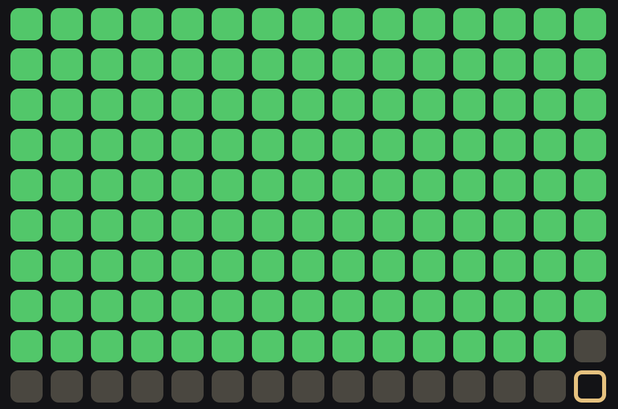
right · 1 waiting · 0 wrong



Batch 3 · 150 complaints

149/150

right · 1 waiting · 0 wrong



267 filed under the right problem

30 set aside: not complaints

2 waiting: not guessed

1 real complaint missed

Sent one by one to the live site. Complaints the system had never seen, scored against a hidden answer key. Made from the same templates as the data, so real complaints will be harder.

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Complaint Compass

65–85%
cheaper

96%
fewer output tokens

2.3×
faster

Thank you. Questions?

Backup

— Accuracy, step by step

Right at every step.

STEP	DONE BY	LIVE TEST	RIGHT
Complaint or not?	Jev	300	99.7%
Which area?	Jev	269	99.6%
Filed by similarity	Free	100	100%
Verified by Jev	Jev	162	100%
Settled by the LLM	Claude Haiku	5 + 2 waiting	100%
Customer found	Free	121 · batch 3	100%

Bars: complaints that reached each step in the live test. 251 of the 300 are worded unlike anything the engine had seen. The engine never sees the answers.

Backup

Live test · the three not filed

Every miss, shown.

Missed

“ande toote hue aaye. Thanks.”

Call · Hinglish · the eggs arrived broken

Set aside as not a complaint. Jev gave it 15%: the “Thanks” read as praise.

Waiting · urgent

“order me cooking oil aaya hi nahi, paise pure kate… Sort it out.”

Chat · Hinglish · an item missing from the order

The LLM was not sure, so it waits for staff instead of a guess. Flagged urgent: money lost.

Waiting

“The detergent had fungus on it.”

Email · an expired product

The LLM was not sure which known problem it is, so it waits for staff.

Waiting is the safe failure: the complaint is stored, visible on the Unconfirmed page, and never filed under the wrong problem.

Ranking

A score you can check with a calculator.

n	×	1–4	×	1–2	×	1–1.25	×	1–1.25
Customers		**Impact**		**Money**		**Anger**		**Trend**
people, not messages		annoyed to harmed		lost money weighs most		kept small: loud never beats hurt		boost when rising

Backup

Customer groups

It shows who is hit hardest.

Students

hit UPI payment timeouts far more often than everyone else

Maharashtra

reports “delivered but not received” more than anywhere else

Rate = customers affected ÷ size of the group, so a big group never looks worst just for being big. Both patterns were planted in the test data, and both show up in the heatmap.

Backup

Any business

One engine. One settings file per business.

Engine

never changes

Grocery — Live
areas, questions, known problems

Telecom — Next
a new file, same engine

Bank — Possible
a new file, same engine

Honesty

What is not done yet.

Limits

- Test data is cleaner than real
- Cut-offs tuned on the same data
- LLM baseline priced, not run
- Live test: 300 complaints, one day

Next

- Telecom pack
- Voice agent demo (webhook built)
- Weekly insights in plain words
- Tune on real labelled data

Backup

The funnel

Only the last few reach the LLM.

